

SOLUTION QUIZ-02 BSE-6B

QUESTION-01

Part	Formula / Logic	Final Answer
A	$RTT_0 + RTT_{Root} + RTT_{TLD} + RTT_{Auth}$	305 ms
B	$RTT_0 + RTT_{TLD} + RTT_{Auth}$	205 ms
C	Time (Part A) + 1 RTT (TCP Handshake)	455 ms

QUESTION-02

1 RTT = 100 ms.

Base HTML Tx = 5 ms.

Image Tx = 0 ms.

TCP Handshake = 1 RTT.

a. Non-Persistent:

- Base HTML: Handshake (100) + Request/Response (100) + Tx (5) = 205 ms.
- 4 Images: $4 \times (100 + 100 + 0) = 800$ ms. **Total = 1005 ms.**

b. Persistent (No Pipeline):

- Handshake (100) + Request/Response Base (100 + 5) = 205 ms.
- Each Image takes: 1 RTT = 100 ms.
- **Total = 205 + 400 = 605 ms.**

c. Persistent (Pipeline):

- Handshake (100) + Request/Response Base (100 + 5) = 205 ms.
- Client parses, sends 4 requests back-to-back.
- The requests travel together; responses travel together. Total time for all 4 images is 1 RTT.
- **Total = 205 + 100 = 305 ms**